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THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



BianLian MAR

Malware Analysis

Classification	TLP:CLEAR
Published	2026-06-13
Version	1.0
Author	Lost Boys Cyber
Report Type	Malware Analysis

TLP:CLEAR | Lost Boys Cyber | BianLian MAR - Malware Analysis

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1. Executive Summary

This report analyzes the Windows x64 sample `de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f` and correlates the file with public reporting on the BianLian intrusion set. Static triage, lightweight reverse-engineering, and local VirtualBox detonation all support the same conclusion: the sample is a Go-based BianLian-associated backdoor / proxy implant rather than the family's historic file-encrypting payload.

The highest-confidence conclusion is not based on one weak string match but on the convergence of multiple evidence classes:

- Go 1.18.3 runtime artefacts and a stripped PE32+ x64 layout.
- Embedded source/build paths referencing `/golang/socks/out/155.94.160.241/2016-7160/client/...`.
- Function and package artefacts including `main.connectForSocks`, `mimux`, and `soso`.
- Local sandbox execution showing `sample.exe` initiate a real outbound TCP connection from `192.168.56.20:58644` to `155.94.160.241:2016`.
- Public BianLian reporting from CISA, Unit 42, and HPE/Juniper describing a custom Go backdoor with SOCKS/tunneling behavior and `mimux` / `soso` modules.

The dynamic run did not show a second-stage payload, ransomware behavior, data staging, or meaningful HTTP/DNS activity attributable to the sample. That absence should not reduce urgency. For this malware family, access tooling is itself the operational danger: it provides covert ingress, relay capability, and an execution foothold that can support hands-on-keyboard activity, lateral movement, credential abuse, and extortion operations.

Field	Assessment
Malware family / cluster	BianLian-associated Go backdoor / proxy implant
Confidence	High for family alignment; medium-high for full operational role in a live intrusion
Platform	Windows x64
Key confirmed IOC	`155.94.160.241:2016`
Static hallmarks	`main.connectForSocks`, `mimux`, `soso`, Go 1.18.3, embedded build paths
Dynamic hallmarks	`sample.exe` launched, no child processes, outbound TCP to `155.94.160.241:2016`, local listening socket on `0.0.0.0:58644`
Most likely role	Post-compromise access, relay, or tunneling implant
Priority action	Hunt for the IOC, isolate impacted hosts, and investigate adjacent credential / remote-access activity

2. Intelligence Requirement

The objective of this analysis was to determine whether the submitted sample is associated with BianLian, characterize its function through static, dynamic, and code-oriented analysis, extract actionable indicators, and provide defensive content suitable for immediate enterprise detection and threat hunting.

This report specifically answers the following questions:

- Is the sample technically consistent with public reporting on BianLian tooling?
- Does the binary behave like an encryptor, a loader, a backdoor, or a tunneling implant?
- What host and network evidence was observed during local sandbox execution?
- What indicators, ATT&CK mappings, detections, and hunt hypotheses should defenders prioritize now?

3. Source Review and Confidence

This assessment combines first-party evidence from the submitted sample and local sandbox execution with public-source enrichment from authoritative reporting.

Source	Type	Relevance	Confidence contribution
Submitted sample archive and extracted PE	Primary evidence	Hashes, file structure, embedded strings, build paths, function artefacts	High
Local TRASHcan / VirtualBox detonation artefacts	Primary evidence	Process execution, TCP connections, DNS / HTTP behavior, generated detection artefacts	High
CISA AA23-136A / 2024 update	Government advisory	Confirms BianLian uses victim-specific Go backdoors, SOCKS5 tunneling, valid-account abuse, and extortion tradecraft	High
HPE / Juniper 2024 activity analysis	Vendor threat research	Describes BianLian Go backdoors using `mimux` and `soso` modules and a hardcoded C2 design	High
Unit 42 threat assessment	Vendor incident-response reporting	Supports BianLian's move from double extortion toward access-and-extortion operations	High
Auto-generated sandbox triage detections	Derived local artefacts	Useful for lead generation, but must be analyst-triaged for false positives	Medium

Assessment caveats:

- The sandbox run confirms execution and one network connection, but it did not surface full post-compromise behavior, likely because the specimen was detonated in an isolated host-only lab with fake services and no real operator infrastructure behind the C2.
- The only behavioral alert generated by the sandbox was an autorun finding for OneDrive under the analyst profile. That artefact is consistent with baseline Windows user activity and is not persuasive evidence of malware persistence in this run.
- Reverse-engineering in this workflow was code-oriented and evidence-driven rather than full decompilation. The conclusions rely on function names, package artefacts, path strings, and runtime behavior rather than a complete control-flow reconstruction.

4. Key Findings

4.1. Sample Metadata

Field	Value
SHA256	`de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f`
SHA1	`e3efe133d1ff69afedb91f359070dbe6d45a5ef6`
MD5	`6564ffc14ab803a961453a118a33830e`

File size	3,489,796 bytes
File type	PE32+ executable for MS Windows 6.01 (GUI), x86-64
Signature status	Unsigned
Build / language indicators	Go runtime artefacts, `go1.18.3`, large `.text` and `.rdata` segments
Sections	`.text`, `.rdata`, `.data`, `.idata`, `.reloc`, `.symtab`
Binary properties	NX enabled, PIC enabled, stripped, overlay present
Build timestamp	`Thu Jan 1 00:00:00 1970` (not operationally meaningful; consistent with Go-linked samples)

4.2. Static and Reverse-Engineering Findings

Finding	Evidence	Assessment
Go-based implant	`rabin2 -I` showed a stripped 64-bit PE with Go-style section layout and runtime strings including `go1.18.3`	Consistent with BianLian public reporting that the actor uses custom Go backdoors
SOCKS / tunneling functionality	String and symbol artefacts included `main.connectForSocks`, `soso.(*Server).ServeConn`, `soso.(*NoAuthAuthenticator).Authenticate`, `soso.(*UserPassAuthenticator).Authenticate`	Strong indicator that the malware provides proxy or relay functionality
Multiplexed session handling	Embedded package/function artefacts for `mimux.Server`, `mimux.(*Session).Accept`, `mimux.(*Session).recvLoop`, and related stream/session handling	Aligns with a backdoor architecture built for multiplexed C2 or pivot traffic
Hardcoded operator infrastructure reference	Raw strings contained `155.94.160.241:2016`	Suggests a compiled-in C2 or relay endpoint rather than purely user-supplied runtime parameters
Embedded developer build paths	Strings showed `/home/admin/prjct/golang/socks/out/155.94.160.241/2016-7160/client/main.go` and adjacent `vendor/mimux/*.go` and `vendor/soso/*.go` files	High-value code-analysis evidence tying the specimen to a specific SOCKS client build tree
No locker-stage artefacts observed	No convincing hits for `.bianlian`, ransom-note artefacts, `vssadmin`, `wbadmin`, or typical encryptor workflow strings	Supports classification as an access implant rather than the final encryption component

The most important code-analysis outcome is that the sample's string and path corpus does not merely resemble generic Go malware. It exposes a developer project structure explicitly oriented around a SOCKS client, includes named internal modules that match public BianLian reporting, and references the same IP/port pair later observed at runtime.

4.3. Dynamic Analysis Findings

The sample was detonated locally in the `win-malware-lab` VirtualBox guest using the TRASHcan local sandbox workflow. After resolving two lab issues during setup (host-only adapter state and an allowed-run-root capture policy), the final detonation completed successfully and produced process, network, and behavioral artefacts.

Sandbox run details:

Field	Value
Run directory	`/home/lost0x01/internal-secops-portal/data/sandbox-runs/2026-06-12_233117_de31a4125eb7`
VM	`win-malware-lab`
Snapshot	`clean-guestadditions-sysmon`
Guest IP	`192.168.56.20`
Host-only gateway / DNS	`192.168.56.1`
Detonated path	`C:\Analysis\Sample\sample.exe`
Sample process ID	`7084`

Observed execution results:

Observation	Evidence	Assessment
Sample executed successfully	`processes.json` and `process_tree_raw.json` show `sample.exe` launched from `C:\Analysis\Sample\sample.exe` under the sandbox PowerShell runner	Confirms the file is runnable and not merely a corrupted or decoy artefact
No child processes spawned	`process_tree_raw.json` showed no descendants from process `7084`	Suggests the specimen did not act as a simple dropper or launcher during this run
Outbound TCP to hardcoded IOC	`tcp_connections.json` captured process `7084` connecting from `192.168.56.20:58644` to `155.94.160.241:2016`	High-confidence behavioral confirmation of the embedded network indicator
Local listening socket present	The same process owned `0.0.0.0:58644` in a listening state before/alongside the remote connection	Consistent with relay, proxy, or local tunnel handling behavior
No malware-attributable DNS or HTTP activity	`analysis.md` reported no suspicious domains, no malware HTTP requests, and no Suricata alerts; observed HTTP requests were background Windows / Microsoft traffic	Supports the hypothesis that the sample prefers direct IP C2 rather than domain-based staging in this run
No convincing persistence established	The only sandbox behavior alert referenced a pre-existing-looking OneDrive autorun path in the analyst user profile	Treat as baseline noise / false positive, not malware-confirmed persistence

Dynamic interpretation:

- The single most important runtime observation is that execution produced the exact network IOC predicted by the static analysis: `155.94.160.241:2016`.
- The presence of a listening socket and an outbound C2 connection is more consistent with a tunneling / backdoor role than with a ransomware locker.
- The absence of follow-on payloads, registry changes, scheduled tasks, or service creation should be interpreted cautiously. A backdoor can still be fully valid even when the lab does not provide a reachable operator workflow or second-stage tasking.

4.4. Classification and ATT&CK Mapping

The total evidence supports a high-confidence classification of this specimen as BianLian-associated access malware, specifically a Go-based backdoor / proxy implant used to establish or maintain command-and-control and facilitate operator access.

ATT&CK Tactic	Technique	Relevance
Command and Control	T1071	Backdoor communications over application-layer protocols remain plausible even though this run showed direct TCP rather than rich HTTP traffic
Command and Control	T1090 / T1090.001 / T1090.002	SOCKS / proxy and relay behavior is strongly indicated by `connectForSocks`, `soso`, and the local listening socket
Ingress Tool Transfer / Remote Services	T1105 / T1219	Public reporting shows BianLian operators pair custom backdoors with remote access tooling
Valid Accounts / External Remote Services	T1078 / T1133	CISA and Unit 42 reporting associate BianLian with stolen credentials, RDP, and VPN exposure
Exploit Public-Facing Application	T1190	CISA reporting links BianLian activity to exploitation paths including ProxyShell
Persistence	T1547.001	Common actor tradecraft, but not confirmed for this specimen in this run
Exfiltration	T1048 / T1567.002	Relevant to broader BianLian operations, though not directly observed in detonation
Impact	T1486	Historically associated with the actor, but not evidenced by this sample

5. Threat Context

BianLian has evolved from a conventional ransomware-and-extortion operation into a cluster that often emphasizes access, data theft, and extortion pressure even when file encryption is absent. That shift matters when evaluating this specimen. Defenders should not dismiss the sample simply because it does not look like a locker.

Public reporting consistently describes the group as:

- Gaining access through exposed remote services, stolen credentials, or exploitable public-facing systems.
- Deploying custom Go backdoors and SOCKS-style tunneling components.
- Maintaining hands-on-keyboard access for reconnaissance, privilege escalation, lateral movement, and extortion preparation.
- Using stealthier access-and-exfiltration tradecraft in cases where encryption is reduced or omitted.

This sample fits that operational profile unusually well. It combines direct evidence of proxy-oriented code structure with a confirmed network connection to the same hardcoded endpoint exposed in its build path and strings. In practical terms, a defender encountering this specimen should assume they are seeing an intrusion-enablement tool that may sit earlier in the kill chain than overt impact actions.

Threat-context question	Assessment
Does the sample fit authoritative BianLian reporting?	Yes, strongly
Is this likely the final ransomware encryptor?	No, the evidence points away from a locker role
Does the absence of encryption reduce risk materially?	No; persistent access tooling is often the more important early-warning signal
Should this be investigated as an isolated malware hit only?	No; treat it as a probable intrusion lead and scope for adjacent actor activity

6. Detection and Hunting

6.1. Immediate Detection Priorities

Detection focus	Why it matters	Example telemetry
`155.94.160.241:2016`	Statically predicted and dynamically confirmed IOC	Firewall logs, EDR network events, Zeek conn logs, proxy telemetry
Processes listening on uncommon local ports and then connecting outbound to the same operator workflow	Matches the observed listener-plus-C2 pattern	EDR socket telemetry, Sysmon Event ID 3, network sensor telemetry
Files or memory containing `mimux`, `soso`, or `main.connectForSocks`	Rare strings with strong family specificity	File scanning, memory scanning, retro-hunts in EDR content indexes
Unsigned Go executables on Windows hosts where they are operationally unusual	Good triage lead for cluster hunting	EDR inventory, software allow-listing exceptions, file prevalence analysis
Suspicious remote-access activity around RDP, VPN, or administrative logons	Consistent with broader BianLian intrusion operations	Authentication logs, Windows security logs, VPN logs, jump-host telemetry

6.2. Indicators of Interest

Type	Value	Confidence	Notes
SHA256	`de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f`	High	Submitted sample hash
SHA1	`e3efe133d1ff69afedb91f359070dbe6d45a5ef6`	High	Static triage
MD5	`6564ffc14ab803a961453a118a33830e`	High	Static triage
IP	`155.94.160.241`	High	Present in strings/build paths and observed at runtime
Port	`2016/TCP`	High	Present in strings/build paths and observed at runtime

Function / string	`main.connectForSocks`	High	Strong behavioral clue for tunneling logic
Package / string	`mimux`	High	Matches public reporting on BianLian tooling
Package / string	`soso`	High	Matches public reporting on BianLian tooling
Build-path fragment	`/golang/socks/out/ 155.94.160.241/2016-7160/ client/main.go`	High	Strong code-analysis artefact

6.3. Sigma Detection Lead

```

title: Potential BianLian Go Backdoor Strings Or Direct IOC
id: 63e02f37-3c75-4d90-a64f-bianlian-mimux-soso-2016
status: experimental
logsource:
  product: windows
  category: file_event
detection:
  selection_hash:
    Hashes|contains:
      - 'de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f'
  selection_name:
    TargetFilename|endswith: '\\sample.exe'
  selection_content:
    FileContent|contains|all:
      - 'mimux'
      - 'soso'
      - 'main.connectForSocks'
  condition: selection_hash or selection_content or selection_name
falsepositives:
  - Malware-analysis labs
  - Internal research repositories containing the specimen
level: high

```

6.4. Microsoft Defender XDR / Sentinel KQL Hunt

```

let sample_sha256 = "de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f";
let c2_ip = "155.94.160.241";
let c2_port = 2016;
let rare_strings = dynamic(["mimux", "soso", "main.connectForSocks", "connectForSocks"]);
DeviceNetworkEvents
| where RemoteIP == c2_ip or (RemoteIP == c2_ip and RemotePort == c2_port)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, LocalPort,
RemoteIP, RemotePort, Protocol, ReportId
| union (
  DeviceFileEvents
  | where SHA256 =~ sample_sha256
  | project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256,
InitiatingProcessFileName, InitiatingProcessCommandLine, ReportId
)
| union (
  DeviceProcessEvents
  | where ProcessCommandLine has_any (rare_strings) or InitiatingProcessCommandLine has_any
(rare_strings)
  | project Timestamp, DeviceName, FileName, ProcessCommandLine, InitiatingProcessFileName,
InitiatingProcessCommandLine, SHA256, ReportId
)
| sort by Timestamp desc

```

6.5. Splunk Hunt

```
(index=sysmon OR index=edr OR index=windows OR index=network)
(
  dest_ip="155.94.160.241"
  OR dest_port="2016"
  OR CommandLine="*connectForSocks*"
  OR CommandLine="*mimux*"
  OR CommandLine="*soso*"
  OR sha256="de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f"
)
| stats min(_time) as first_seen max(_time) as last_seen values(host) as hosts values(user) as users
values(Image) as images values(CommandLine) as cmd values(dest_ip) as dest_ips values(dest_port) as
dest_ports by sha256 parent_process process_name
| convert ctime(first_seen) ctime(last_seen)
```

6.6. YARA Rule

```
rule WIN_BianLian_Go_Backdoor_Mimux_Soso_ConfirmedC2
{
  meta:
    author = "Lost Boys Cyber"
    description = "Detects a BianLian-associated Go backdoor using mimux/soso artefacts and confirmed
C2 indicator"
    sha256 = "de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f"
    date = "2026-06-12"
  strings:
    $mz = { 4D 5A }
    $s1 = "mimux" ascii
    $s2 = "soso" ascii
    $s3 = "main.connectForSocks" ascii
    $s4 = "155.94.160.241" ascii
    $s5 = "/golang/socks/out/155.94.160.241/2016-7160/client/main.go" ascii
  condition:
    $mz at 0 and 3 of ($s1,$s2,$s3,$s4,$s5)
}
```

6.7. Hunt Hypotheses

- Hosts that contacted `155.94.160.241:2016` may also show valid-account abuse, remote desktop enablement, or unusual administrative session activity in the same window.
- Endpoints containing this specimen may have been used as relay points rather than encryption victims, so network pivoting and credential movement deserve equal priority to malware eradication.
- Additional BianLian tooling may exist on affected systems even when this sample itself does not create persistence or drop secondary binaries during lab execution.
- Because the sample uses a direct IP at runtime, detections focused only on suspicious domains may miss comparable variants.

7. Recommendations

Priority	Recommendation	Rationale
Immediate	Block and alert on `155.94.160.241` and specifically review historical flows to `2016/TCP`	This IOC was both embedded and behaviorally confirmed
Immediate	Quarantine any host containing SHA256 `de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f`	Prevent reuse of a confirmed access implant
Immediate	Scope authentication and remote-access telemetry for the affected host(s)	BianLian frequently relies on valid accounts, RDP, VPN access, and hands-on-

		keyboard follow-on activity
High	Hunt for unsigned Go executables with `mimux`, `soso`, or `connectForSocks` artefacts across the estate	Strong family-specific pivot for retro-hunting
High	Investigate hosts for listener behavior paired with outbound operator traffic	Matches the local-listener plus C2-connection pattern seen in detonation
High	Review adjacent systems for staging, exfiltration, remote tooling, and lateral-movement artefacts	The specimen appears positioned earlier than the final impact phase
Medium	Preserve the specimen and sandbox artefacts for deeper reversing if this case escalates	Full decompilation could recover configuration nuances, error handling, and operator tradecraft
Medium	Add direct-IP network detections alongside domain-based controls	This run showed no malware-attributable DNS activity before C2 contact

Operational note: the absence of ransom-note or encryption artefacts should not be treated as exculpatory. In BianLian cases, early access tooling may be the most valuable detection opportunity defenders get before extortion pressure becomes visible.

8. References

- CISA, `#StopRansomware: BianLian Ransomware Group`, AA23-136A, <https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-136a>
- CISA / FBI / ASD ACSC, `#StopRansomware: BianLian Data Extortion Group` PDF update, <https://www.cisa.gov/sites/default/files/2024-11/aa23-136a-joint-csa-stopransomware-bianlian-ransomware-group.pdf>
- Palo Alto Networks Unit 42, `Threat Assessment: BianLian`, <https://unit42.paloaltonetworks.com/bianlian-ransomware-group-threat-assessment/>
- HPE Threat Research, `BianLian Ransomware Group: 2024 Activity Analysis`, <https://community.hpe.com/t5/hpe-threat-labs/bianlian-ransomware-group-2024-activity-analysis/ba-p/7266394>
- Submitted evidence archive: `/home/lost0x01/internal-secops-portal/data/uploads/20260612\_225918\_549706/de31a4125eb74d0b7cbf2451b40fdb2d66d279a8b8fd42191660b196a9ac468f.7z`
- Successful local sandbox run directory: `/home/lost0x01/internal-secops-portal/data/sandbox-runs/2026-06-12\_233117\_de31a4125eb7`
- Key local artefacts reviewed in this workflow: `analysis.md`, `summary.json`, `behavior\_summary.json`, `guest\_artifacts/process\_tree\_raw.json`, `guest\_artifacts/processes.json`, `guest\_artifacts/tcp\_connections.json`, `static\_triage.json`, `rule.yar`, and `sigma\_behavior.yml`